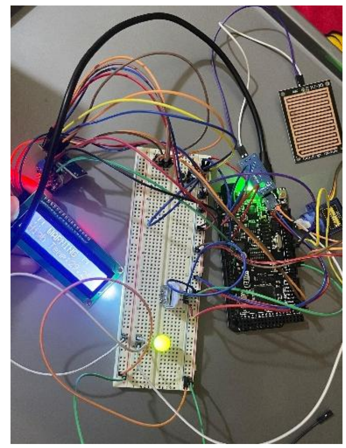
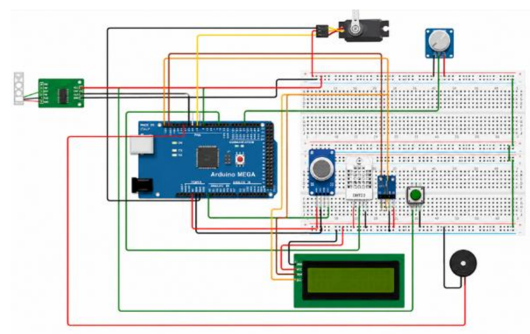
Serena

Hardware Wiring, Schematics, Demonstration Videos, and Calibration



Serena was mainly responsible for the electrical wiring and hardware interconnection of the Smart Weather Station. She connected the sensors, actuators, and user-interface components to the Arduino Mega according to the final pin assignments. Her work included connecting the DHT22, BH1750, BMP280, MQ135, HX711, load cell, water-flow sensor, LCD, push button, servo motor, and buzzer. She also checked that all modules shared a common ground and that the LCD, BH1750, and BMP280 were correctly connected to the shared I^2C bus.



Serena prepared and reviewed the complete wiring schematic used in the hardware-design section of the report. She verified the signal-pin assignments, power connections, interrupt connection for the water-flow sensor, and the SDA and SCL connections used by the I^2C devices. This responsibility was important because the project requirements specify that the wiring must be documented using WOKWI.

She also participated in hardware troubleshooting during system integration. This included checking loose wires, testing component power connections, confirming the operation of the LCD and button, and helping identify hardware-related causes when sensors did not provide valid readings.

Serena was additionally responsible for recording and organizing the demonstration videos. She documented the final prototype, LCD screens, servo movement, buzzer warning, load-cell operation, water-flow measurement, and complete system operation. These videos provided evidence that the prototype could acquire multiple sensor values, display system status, and produce automatic responses.

Calibration was a shared responsibility between Serena and Malak. Serena prepared the physical calibration setups, connected the sensors, positioned known reference loads, arranged the flow-sensor tubing and measuring container, and recorded raw readings during the tests. She also helped verify the servo positions and light thresholds during BH1750 testing.

Personal reflection:

Through this project, I improved my understanding of practical sensor wiring, shared communication buses, interrupt-based inputs, and hardware troubleshooting. I learned that correct wiring requires more than following a pin table because grounding, power consumption, loose connections, and component placement can strongly affect system reliability. I also gained experience preparing clear schematics, documenting a working prototype, and performing calibration tests using known references.